DP Computer Science: A 4.2.3 Dimensionality Reduction

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1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Describe the **"Curse of Dimensionality"** and its negative impacts on machine learning models
- Explain the **importance of dimensionality reduction** in improving model performance and efficiency
- Identify and describe considerations related to the curse of dimensionality, including **overfitting, computational complexity, and data sparsity**
- Conceptually understand that dimensionality reduction aims to **reduce the number of variables while preserving the most important information**

Key Vocabulary

Term	Definition
Dimensionality	The number of features (attributes) in a dataset
Curse of Dimensionality	The problems that arise when working with high-dimensional data
Data sparsity	The phenomenon where data points become spread out and isolated as dimensions increase
Overfitting	A model that learns noise in the training data instead of true patterns
Computational complexity	The amount of computation required as the number of features increases
Distance metric	A measure of similarity or dissimilarity between data points (e.g., Euclidean distance)

Term	Definition
Dimensionality reduction	The process of reducing the number of features while preserving essential information
Feature Selection	Keeping or discarding original features
Feature Extraction	Transforming original features into a smaller set of new features

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Approaches to Learning (ATL) Elements

Skill	Description
Thinking (Abstraction & Critical Thinking)	Grasping the abstract concept of high-dimensional space; critically analysing why adding more features can harm model performance
Communication	Using analogies and simplified examples to explain complex ideas like data sparsity and the curse of dimensionality

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2. The Curse of Dimensionality

What is the Curse of Dimensionality?

> **The "Curse of Dimensionality" refers to the problems that arise when working with high-dimensional data. As the number of features increases, the data becomes harder to work with and models can actually perform worse.**

The Core Problem

```text
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```
┌─────────────────────────────────────────────────────────────────┐
│              THE CURSE OF DIMENSIONALITY                        │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    As the number of features increases...                      │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │  DATA BECOMES SPARSE (Empty)                            │  │
│    │  Points are far apart; patterns are hard to find        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │  COMPUTATIONAL COST INCREASES                           │  │
│    │  More calculations = slower training and inference      │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │  OVERFITTING BECOMES MORE LIKELY                       │  │
│    │  The model learns noise instead of real patterns        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │  DISTANCE METRICS BECOME LESS USEFUL                   │  │
│    │  In high dimensions, everything seems equally far      │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │  VISUALISATION BECOMES IMPOSSIBLE                       │  │
│    │  We cannot visualise data with >3 dimensions            │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
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```

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3. The Crowded Room vs. Empty Warehouse Analogy

Understanding Data Sparsity

```
text
```

```
┌─────────────────────────────────────────────────────────────────┐
│          CROWDED ROOM vs. EMPTY WAREHOUSE ANALOGY              │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌──────────────────────────────────────────────────────────┐ │
│    │              1D: A LINE                                 │ │
│    │                                                          │ │
│    │    ●──●────●──●────●──●────●──●────●──●                │ │
│    │                                                          │ │
│    │    10 dots on a line → relatively close together        │ │
│    │    Dense space → patterns are easy to see               │ │
│    └──────────────────────────────────────────────────────────┘ │
│                                                                 │
│    ┌──────────────────────────────────────────────────────────┐ │
│    │              2D: A SQUARE                               │ │
│    │                                                          │ │
│    │    ●     ●  ●                                            │ │
│    │                ●                                         │ │
│    │          ●             ●                                 │ │
│    │    ●               ●                                     │ │
│    │         ●        ●                                       │ │
│    │                                                          │ │
│    │    10 dots on a square → more spread out                │ │
│    │    Sparser space → patterns are harder to see           │ │
│    └──────────────────────────────────────────────────────────┘ │
│                                                                 │
│    ┌──────────────────────────────────────────────────────────┐ │
│    │              3D: A CUBE                                │ │
│    │                                                          │ │
│    │    The 10 dots are spread across a volume               │ │
│    │    Very sparse space → patterns are extremely          │ │
│    │    difficult to see                                      │ │
│    └──────────────────────────────────────────────────────────┘ │
│                                                                 │
│    As dimensions increase, the same number of points          │
│    becomes spread out and isolated.                           │
│                                                                 │
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```

Key Insight

> *"In a high-dimensional space, data points become like ships in a vast ocean—far apart and isolated. Finding patterns or 'neighbours' becomes extremely difficult."*

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4. Consequences of the Curse of Dimensionality

4.1 Data Sparsity

Aspect	Description
Definition	Data points become spread out and isolated as dimensions increase
Problem	Patterns become harder to detect
Impact	Models require exponentially more data to find meaningful patterns

Visual Example:

Dimensions	Points	Density
1D (Line)	10	Dense
2D (Square)	10	Sparse
3D (Cube)	10	Very Sparse
10D	10	Extremely Sparse

4.2 Overfitting

Aspect	Description
Definition	A model learns noise in the training data instead of true patterns

Aspect	Description
Why It Happens	High-dimensional data has many ways to fit noise
Impact	Model performs well on training data but poorly on new data

```text
┌─────────────────────────────────────────────────────────────────┐
│              OVERFITTING EXAMPLE                              │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              TRAINING DATA                            │  │
│    │                                                         │  │
│    │    •    •     •                                       │  │
│    │       •   •      •                                    │  │
│    │    •     • •     •                                    │  │
│    │                                                         │  │
│    │    Complex model fits EVERY point perfectly!          │  │
│    │    But it's just memorising the training data.        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                               │                                 │
│                               ▼                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              NEW DATA                                 │  │
│    │                                                         │  │
│    │    •    •     •                                       │  │
│    │       •   •      •                                    │  │
│    │    •     • •     •                                    │  │
│    │                                                         │  │
│    │    The complex model performs poorly!                 │  │
│    │    It learned the noise, not the real pattern.        │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

4.3 Computational Complexity

Aspect	Description
Definition	More features require more calculations
Why It Happens	Many algorithms scale with the number of features
Impact	Training becomes slower; models are less efficient

Example:

Features	Calculation Time
10 features	10 units of time
100 features	100 units of time
1,000 features	1,000 units of time

4.4 Distance Metrics Become Less Useful

Aspect	Description
Definition	In high dimensions, all points seem equally far from each other
Why It Happens	The volume of space grows exponentially
Impact	Distance-based algorithms (k-NN) perform poorly

Example:

```python
# In 1D:
Distance between 0 and 10 = 10 units
Distance between 0 and 20 = 20 units

# In 100D:
The average distance between any two points is approximately the same!
```

4.5 Visualisation Challenges

Aspect	Description
Definition	We cannot visualise data with more than 3 dimensions
Why It Happens	Human perception is limited to 3D space
Impact	Exploratory data analysis becomes difficult

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5. Dimensionality Reduction as the Solution

What is Dimensionality Reduction?

> **Dimensionality reduction is the process of reducing the number of features in a dataset while preserving as much important information as possible.**

Feature Selection vs. Dimensionality Reduction

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┌─────────────────────────────────────────────────────────────────┐
│        FEATURE SELECTION vs. DIMENSIONALITY REDUCTION          │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ┌─────────────────────────────────────────────────────────┐  │
│    │              FEATURE SELECTION                         │  │
│    │                                                         │  │
│    │    • Keeps OR discards original features               │  │
│    │    • Selects a subset of the original features         │  │
│    │    • Features remain the same                          │  │
│    │    • Example: Removing irrelevant columns              │  │
│    │                                                         │  │
│    │    [Age, Height, Weight, Salary]                      │  │
│    │         │                 │                             │  │
│    │         ▼                 ▼                             │  │
│    │    [Age, Height, Weight]                            │  │
│    └─────────────────────────────────────────────────────────┘  │
│                                                                 │
```

```
│    ┌──────────────────────────────────────────────────────────┐   │
│    │            DIMENSIONALITY REDUCTION                   │   │
│    │                                                          │   │
│    │    • Transforms original features into NEW features   │   │
│    │    • Combines features to create new features         │   │
│    │    • New features are combinations of originals       │   │
│    │    • Example: PCA creates principal components        │   │
│    │                                                          │   │
│    │    [Age, Height, Weight, Salary]                      │   │
│    │          │                 │                             │   │
│    │          ▼                 ▼                             │   │
│    │    [PC1, PC2]  (Combinations of original features)   │   │
│    └──────────────────────────────────────────────────────────┘   │
│                                                                   │
└───────────────────────────────────────────────────────────────────┘
```

The Soup vs. Stock Analogy

```text
┌───────────────────────────────────────────────────────────────────┐
│              SOUP vs. STOCK ANALOGY                            │
├───────────────────────────────────────────────────────────────────┤
│                                                                   │
│    ┌──────────────────────────────────────────────────────────┐   │
│    │              FEATURE SELECTION = SOUP                 │   │
│    │                                                          │   │
│    │    • Remove some ingredients                            │   │
│    │    • What remains is still the same ingredients       │   │
│    │    • Example: Removing onions from the soup           │   │
│    └──────────────────────────────────────────────────────────┘   │
│                                                                   │
│    ┌──────────────────────────────────────────────────────────┐   │
│    │          DIMENSIONALITY REDUCTION = STOCK             │   │
│    │                                                          │   │
│    │    • Simmer ingredients to create a new concentrate   │   │
│    │    • New product is a combination of all ingredients  │   │
│    │    • Example: Boiling vegetables to make stock        │   │
│    └──────────────────────────────────────────────────────────┘   │
```

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│                                                    │
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```

6. Why Dimensionality Reduction Matters

Benefits of Dimensionality Reduction

Benefit	Explanation
Reduces Overfitting	Fewer features = less chance to learn noise
Improves Efficiency	Less computation = faster training
Reduces Memory Usage	Less data = less storage
Improves Visualisation	Can reduce to 2D or 3D for visualisation
Improves Performance	Distance metrics work better in lower dimensions
Reduces Noise	Removes irrelevant features

The Goal

> *"Dimensionality reduction aims to reduce the number of variables while preserving the most important information."*

7. Conceptual Activity: Visualising Sparsity

Instructions

1. Draw a **1D line** and place **10 dots** randomly on it
2. Draw a **2D square** of the same side length and place **10 dots** randomly in it
3. Imagine a **3D cube** with **10 dots** randomly placed inside it

Discussion Questions

Question	Purpose
"In which space are the dots closest to each other?"	Understand density
"What happens to the average distance between points as we add dimensions?"	Understand sparsity
"How does this make finding 'neighbours' or 'patterns' harder?"	Understand the curse

Expected Observations

Dimension	Density	Pattern Detection
1D	High	Easy
2D	Medium	Moderate
3D	Low	Difficult
High Dimensions	Very Low	Extremely Difficult

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8. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Recap & Hook	10 min	Recap data cleaning and feature selection; introduce the curse of dimensionality
Lecture: Curse of Dimensionality	20 min	Data sparsity, overfitting, computational complexity, distance

Phase	Time	Activity
		metrics, visualisation
Conceptual Activity	15 min	"Visualising Sparsity" — drawing dots in 1D, 2D, and 3D
Lecture: Dimensionality Reduction	10 min	Introduction to dimensionality reduction; feature selection vs. dimensionality reduction
Consolidation	5 min	Summary; preview next lesson (Linear Regression)

Discussion Questions

Question	Purpose
"Why can adding more features sometimes make a model perform worse?"	Understand overfitting
"What is data sparsity?"	Understand the core concept
"Why does computational complexity increase with more features?"	Understand efficiency
"What is the difference between feature selection and dimensionality reduction?"	Distinguish concepts

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9. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions

Assessment Type	Description
Conceptual Activity	Assess understanding of sparsity
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the curse of dimensionality?
2. What is data sparsity?
3. Why does overfitting become more likely with high-dimensional data?
4. What is the difference between feature selection and dimensionality reduction?
5. Give one benefit of dimensionality reduction.

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10. Differentiation

Level	Strategy
Support	Use analogies and visual aids; step through the "Visualising Sparsity" activity carefully
Challenge	Research techniques like PCA and t-SNE; explore real-world applications of dimensionality reduction

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11. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.2.3 with examples)
- **Worksheet:** Dimensionality reduction concepts
- **Worksheet Answers**
- **Quizlet** (A4.2.3 vocabulary flashcards)

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12. Summary: Key Takeaways

Concept	Key Point
Curse of Dimensionality	High-dimensional data causes problems
Data Sparsity	Points become isolated; patterns are hard to find
Overfitting	Model learns noise instead of patterns
Computational Complexity	More features = slower training
Distance Metrics	Become less useful in high dimensions
Visualisation	Cannot visualise >3 dimensions
Dimensionality Reduction	Reduces features while preserving information
Feature Selection	Keeps/discards original features
Feature Extraction	Creates new features from originals

```text
┌─────────────────────────────────────────────────────────────────┐
│                    KEY REMINDERS                                │
├─────────────────────────────────────────────────────────────────┤
│                                                                 │
│    ✅ More features is NOT always better                        │
│    ✅ High dimensions = data sparsity                           │
│    ✅ High dimensions = overfitting risk                        │
│    ✅ High dimensions = computational cost                      │
│    ✅ Feature Selection = choose from originals                 │
│    ✅ Dimensionality Reduction = create new features            │
│    ✅ The goal = reduce features, preserve information          │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

> *"The curse of dimensionality shows us that adding more features can actually harm our models—dimensionality reduction is the cure."*